Equalizer Lowpass FIR Filter Design via Convex Optimisation

1 Overview

A complex-valued lowpass FIR filter is designed whose passband target matches the UCDC equalizer frequency response. The filter is found by solving a second-order cone programme (SOCP) using CVXPY.

Signal chain parameters

Parameter	Value
Sample rate f_s	750 MHz (two-sided, Nyquist = $\pm$375 MHz)
Passband edge f_p	$\pm$162.5 MHz
Stopband edge f_{stop}	$\pm$250.0 MHz
Filter taps N	63
Passband tolerance δ_p	0.05
Stopband tolerance δ_s	0.1

2 Optimisation Formulation

The DTFT of the complex tap vector $\mathbf{h} \in \mathbf{C}^N$ is

$$H(\omega) = \sum_{n=0}^{N-1} h[n]\, e^{-j\omega n} = \mathbf{a}(\omega)^\top \mathbf{h}, \quad \mathbf{a}(\omega) = \left[1,\, e^{-j\omega},\, \ldots,\, e^{-j\omega(N-1)}\right]^\top.$$

Sampling over a frequency grid turns this into the linear map $\mathbf{H} = \mathbf{A}\mathbf{h}$, where $A_{kn} = e^{-j\omega_k n}$.

Let $\mathcal{P}$, $\mathcal{S}$, and $\mathcal{T}$ denote the index sets of the passband, stopband, and transition-band frequency grid points, respectively. The **fixed-tolerance** problem is

$$\begin{aligned}
\underset{\mathbf{h}}{\text{minimise}} \quad & \sum_{k\in\mathcal{P}} |\mathbf{a}(\omega_k)^\top \mathbf{h} - H_d(\omega_k)| + \lambda \|\mathbf{h}\|^2 \\
\text{subject to} \quad & |\mathbf{a}(\omega_k)^\top \mathbf{h} - H_d(\omega_k)| \le \delta_p, \quad k \in \mathcal{P} \\
& |\mathbf{a}(\omega_k)^\top \mathbf{h}| \le \delta_s, \quad k \in \mathcal{S} \\
& |\mathbf{a}(\omega_k)^\top \mathbf{h}| \le \delta_{t,k}, \quad k \in \mathcal{T}.
\end{aligned}$$

The transition-band bound $\delta_{t,k}$ decreases linearly from $\max_{k\in\mathcal{P}} |H_d(\omega_k)|$ at the passband edge to $3\delta_s$ at the stopband edge, preventing large overshoot in the unconstrained gap while keeping the SOCP feasible.

3 Filter Response

Figure 1 shows the designed filter's magnitude (dB) and phase response overlaid with the desired equalizer response $H_d(\omega)$.

Figure 2 shows the in-band magnitude response zoomed to $\pm$5 MHz beyond the passband edges, with the y-axis limited to $[-0.9, +0.1]$ dB to resolve the passband ripple.

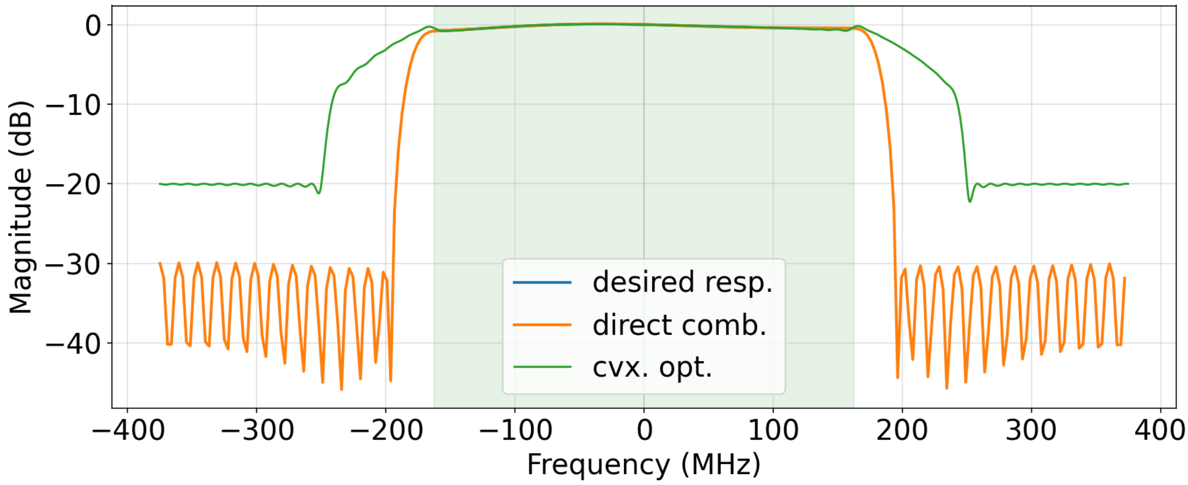

Figure 1: Magnitude (top) and phase (bottom) of the designed equalizer lowpass FIR filter. The green shaded regions indicate the passband ±162.5 MHz. The dashed red line is the desired response $H_d(\omega)$ interpolated from the 256-bin UCDC equalizer FFT.

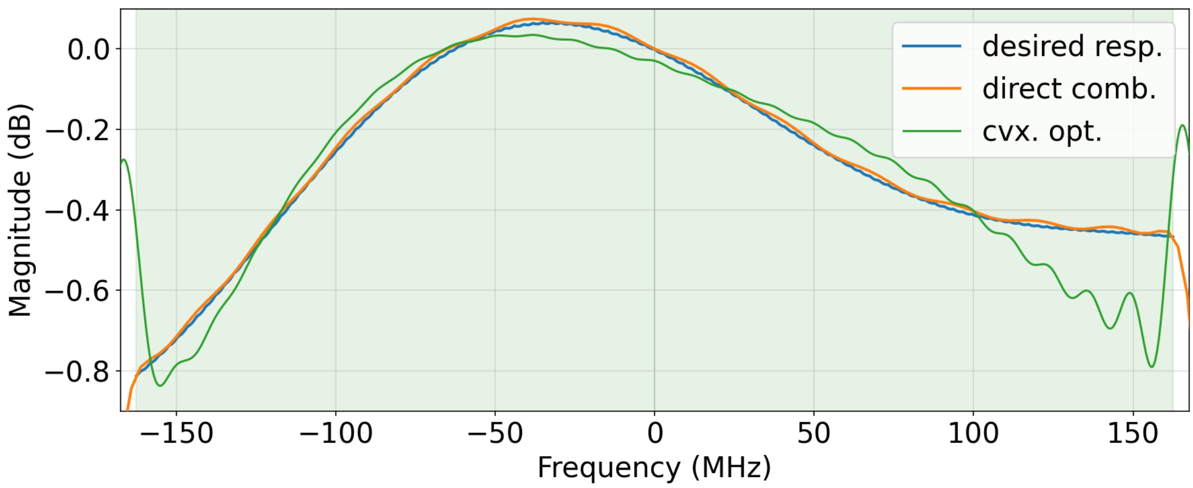

Figure 2: In-band magnitude response zoomed to $[-167.5, +167.5]$ MHz with a $[-0.9, +0.1]$ dB y-axis. The green shaded regions indicate the passband ±162.5 MHz.

4 Comparison: CVX Fixed vs. Remez IRLS

Figure 3 compares the desired frequency response with the two lowpass FIR designs: the CVXPY fixed-tolerance solver (no transition-band constraint) and the complex Remez algorithm (Lawson IRLS). Both designs share identical passband and stopband grids and the same DC-normalised UCDC equalizer target. The top panel shows the full-band magnitude response; the bottom panel zooms into the passband region with the y-axis limited to $[-1, +0.2]$ dB to resolve the in-band ripple.

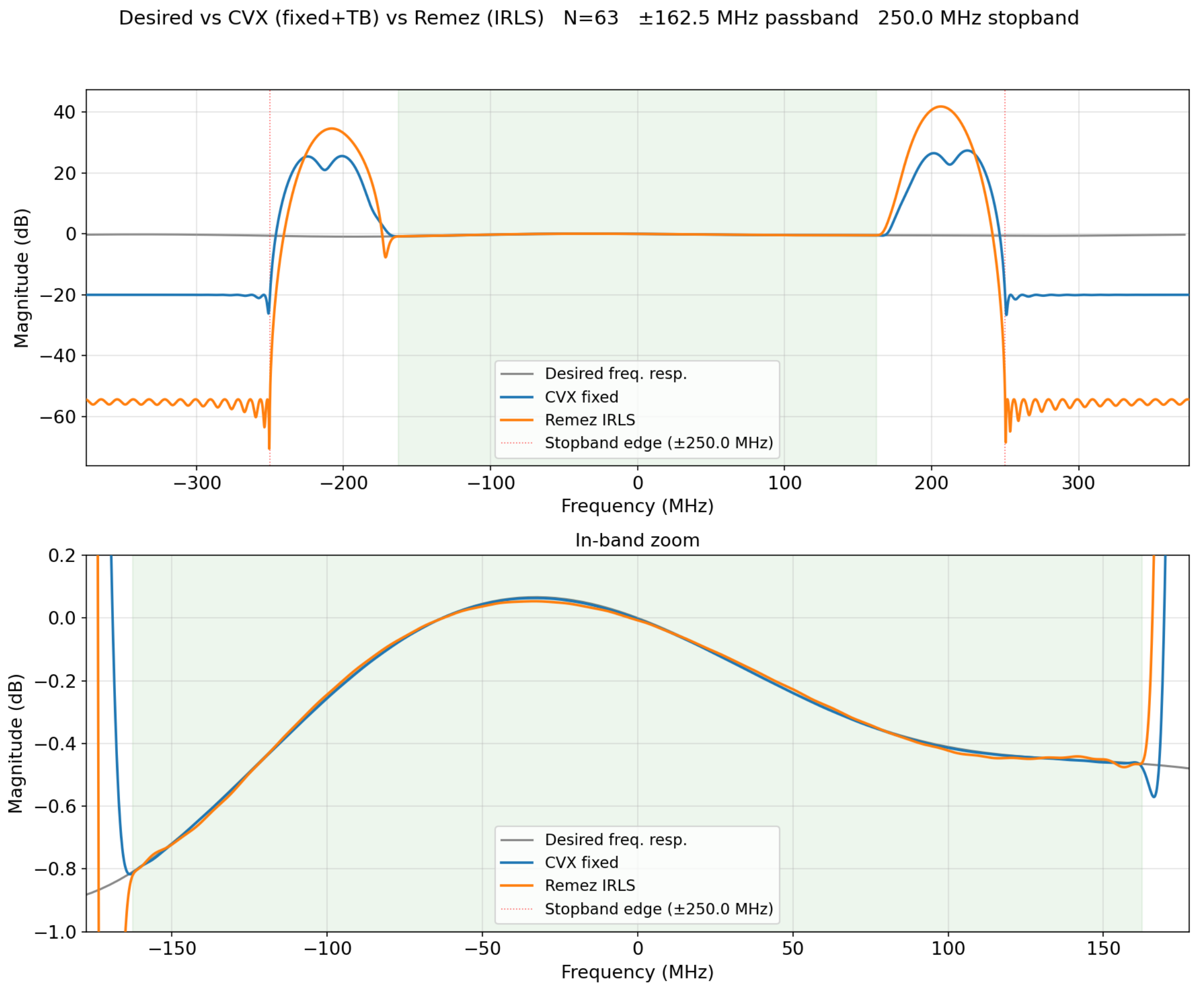

Figure 3: Overlay of the desired equalizer response, the CVX fixed-tolerance design, and the Remez IRLS design ($N = 63$, ± 162.5 MHz passband, 250 MHz stopband). Top: full-band magnitude (dB). Bottom: in-band zoom with y-axis $[-1, +0.2]$ dB. The green shaded region indicates the passband; dotted red lines mark the stopband edges.